

1. Agnostic PAC Learning

1. Motivation

Realizable assumption ($\exists h^* \in \mathcal{H}$ with $R(h^*) = 0$) is often unrealistic.

Agnostic setting: No assumption that perfect hypothesis exists. Allow $\inf_{h \in \mathcal{H}} R(h) > 0$.

Goal: Find h close to best in class:

$$R(h) \leq \inf_{h' \in \mathcal{H}} R(h') + \varepsilon$$

2. Empirical Risk Minimization

ERM Algorithm:

$$h_S = \arg \min_{h \in \mathcal{H}} R_{\text{emp}}(h)$$

Key question: When does R_{emp} approximate R **uniformly** over all $h \in \mathcal{H}$?

Uniform convergence: For all $h \in \mathcal{H}$ simultaneously,

$$|R(h) - R_{\text{emp}}(h)| \leq \varepsilon$$

2. Uniform Convergence

1. Core Concept

Definition: $\mathcal{H}$ has **uniform convergence** property if:

$$\mathbb{P}_S \left[\sup_{h \in \mathcal{H}} |R(h) - R_{\text{emp}}(h)| > \varepsilon \right] \rightarrow 0$$

as $m \rightarrow \infty$.

Implication: Sample average errors converge to true errors uniformly over entire hypothesis class.

2. Finite Hypothesis Sets

Agnostic bound:

$$\mathbb{P} \left[\sup_{h \in \mathcal{H}} |R(h) - R_{\text{emp}}(h)| > \varepsilon \right] \leq 2|\mathcal{H}| \exp(-2m\varepsilon^2)$$

Sample complexity:

$$m \geq \frac{2}{\varepsilon^2} \left(\log |\mathcal{H}| + \log \frac{2}{\delta} \right)$$

Note: $1/\varepsilon^2$ (vs $1/\varepsilon$ in realizable).

3. Fundamental Theorem

1. Statement

For binary classification, the following are equivalent:

1. $\mathcal{H}$ is PAC-learnable
2. $\mathcal{H}$ has uniform convergence
3. $\text{VCdim}(\mathcal{H}) < \infty$

Key insight: Finite VC dimension is necessary and sufficient for learnability.

2. Implications

From finite to infinite: VC dimension extends PAC framework to infinite hypothesis classes.

Sample complexity: $m = O\left(\frac{d}{\varepsilon^2}\right)$ where $d = \text{VCdim}(\mathcal{H})$.

No Free Lunch: If $\text{VCdim}(\mathcal{H}) = \infty$, then $\mathcal{H}$ is not PAC-learnable.

4. Symmetrization

1. Ghost Samples

Technique: Introduce "ghost" sample $S' = \{(x'_1, y'_1), \dots, (x'_m, y'_m)\}$ drawn independently from same distribution.

Key identity:

$$\mathbb{E}_S \left[\sup_{h \in \mathcal{H}} (R(h) - R_{\text{emp}}(h)) \right] \leq 2\mathbb{E}_{S, S'} \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \sigma_i (h(x_i) - h(x'_i)) \right]$$

where $\sigma_i \in \{-1, +1\}$ are Rademacher random variables.

2. Rademacher Complexity

Definition:

$$\hat{\mathcal{R}}_m(\mathcal{H}) = \mathbb{E}_\sigma \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \sigma_i h(x_i) \right]$$

Measures: Ability of $\mathcal{H}$ to fit random noise.

Generalization bound:

$$R(h) \leq R_{\text{emp}}(h) + 2\hat{\mathcal{R}}_m(\mathcal{H}) + O\left(\sqrt{\frac{\log(1/\delta)}{m}}\right)$$

5. Sample Complexity

1. Upper Bounds

Finite VC dimension d :

$$m = O\left(\frac{d \log(m/d) + \log(1/\delta)}{\varepsilon^2}\right)$$

Simplified (ignoring log factors):

$$m = \Theta\left(\frac{d}{\varepsilon^2}\right)$$

Interpretation: Sample size linear in VC dimension.

2. Lower Bounds

Matching lower bound: For any learning algorithm and hypothesis class with $\text{VCdim}(\mathcal{H}) = d$,

$$m = \Omega\left(\frac{d}{\varepsilon^2}\right)$$

Conclusion: VC-based bounds are tight up to log factors.

6. No Free Lunch

1. Statement

No universal learner: There is no learning algorithm that PAC-learns every hypothesis class.

Proof idea: For any algorithm A , can construct a "hard" distribution where A fails with high probability.

Implication: Must make assumptions about target concept (restrict to $\mathcal{H}$ with finite VC dimension).

2. Interpretation

Inductive bias required: Learning requires assumptions. Can't learn everything from data alone.

VC dimension measures bias: Larger d = less restrictive assumption = more samples needed.

7. Structural Risk Minimization

1. Motivation

Trade-off: Larger $\mathcal{H}$ can fit data better but requires more samples.

Idea: Balance empirical risk against hypothesis class complexity.

SRM principle: Choose from nested classes $\mathcal{H}_1 \subset \mathcal{H}_2 \subset \dots$ to minimize:

$$R_{\text{emp}}(h) + \text{complexity-penalty}(\mathcal{H}_n)$$

8. Key Pitfalls

Uniform vs pointwise: Uniform convergence is over **all** $h \in \mathcal{H}$ simultaneously. Pointwise convergence (for fixed h) is weaker.

Agnostic $\neq$ noisy: "Agnostic" means no perfect h in $\mathcal{H}$, not that labels are noisy. Can have deterministic labels and still be agnostic.

VC dimension = infinite: Means class is **not PAC-learnable**, regardless of algorithm. No amount of data guarantees learning.

ERM computability: ERM may be NP-hard to compute even when class is PAC-learnable. Learnability $\neq$ efficient learnability.

Symmetrization doubles sample: Ghost sample trick introduces factor of 2 in bounds.